Introduction to Information Theory

The ideas behind every phone call, ZIP file and AI model

Part A · Measuring information

probability and logs, information, surprise and the bit, entropy, shared information

Part B · Using information

compression, noisy channels, cross-entropy and KL, information theory in AI

Ibrahim Sobh, PhD · ibrahimsobh.github.io/introduction-to-information-theory

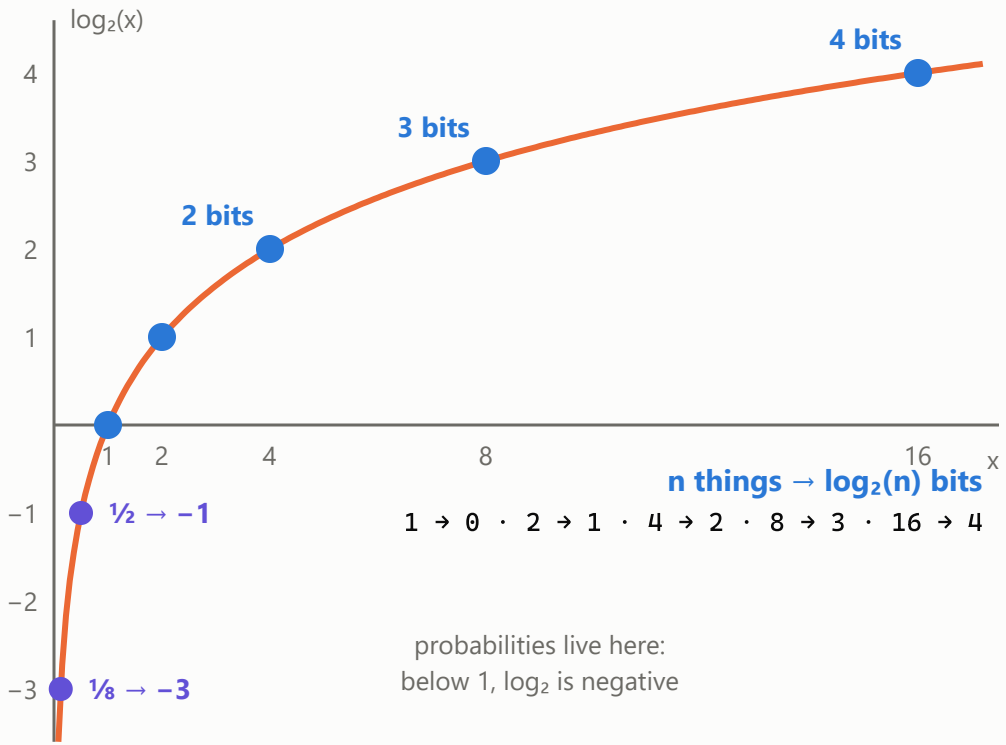
PART A

Measuring information

1 Probability and logs
2 What is information?
3 Surprise and the bit
4 Entropy
5 Shared information

Logs count yes/no answers

- A **probability** is a number from 0 (never) to 1 (always); all outcomes add up to 1; independent events multiply
- **$2^b = n \Leftrightarrow \log_2(n) = b$**: b bits can write n codes, and n things need $\log_2(n)$ bits
- $\log_2$ counts halvings: 16 → 8 → 4 → 2 → 1 is 4 steps, so $\log_2(16) = 4$
- For probabilities (below 1), $\log_2$ is negative; logs turn × into +

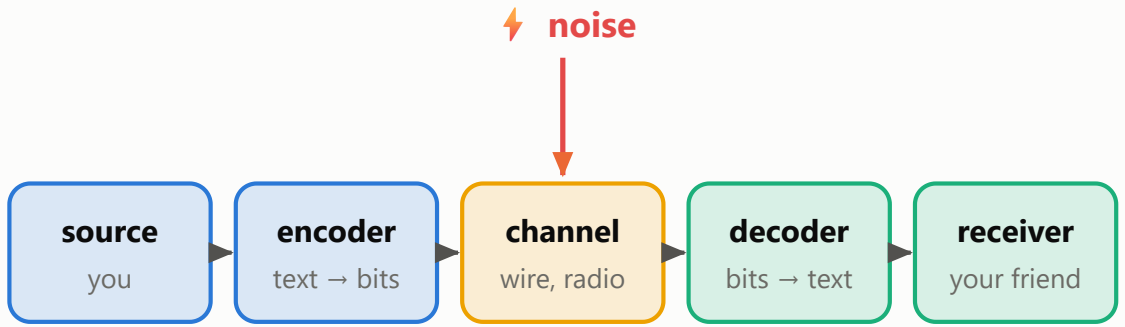


KEY INSIGHT

$\log_2(n)$ is the number of fair yes/no answers needed to pick one of n things.

Information is what reduces uncertainty

- Claude Shannon, 1948: information is what you **didn't know before**, not what a message means
- Every communication: **source** → **encoder** → **channel (noise)** → **decoder** → **receiver**
- Shannon's two big questions: how small can a message be (**compression**)? How fast can it travel through noise (**capacity**)?



sent: `0 1 0 0 1 0 0 0` = "H" (ASCII 72)

received: `0 1 0 1 1 0 0 0` one bit flipped → "X"

The same five parts in a phone call, Wi-Fi, USB and a space probe.

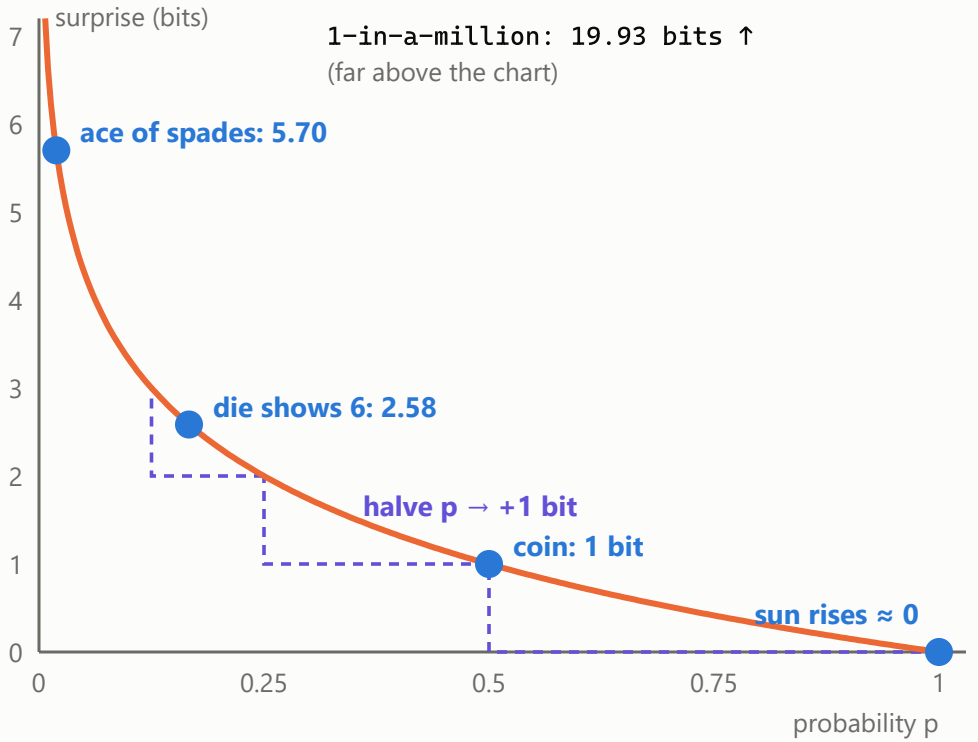
KEY INSIGHT

"The sun rose" tells you almost nothing; "it snowed in the desert" tells you a lot. Information measures surprise, not meaning.

Rare events carry more bits

The **surprise** (self-information) of an event with probability p is $-\log_2(p)$ bits.

Event	p	Surprise
The sun rises	≈ 1	≈ 0 bits
A coin lands heads	1/2	1 bit
A die shows 6	1/6	2.58 bits
Ace of spades	1/52	5.70 bits
Win a 1-in-a-million draw	1/1,000,000	19.93 bits

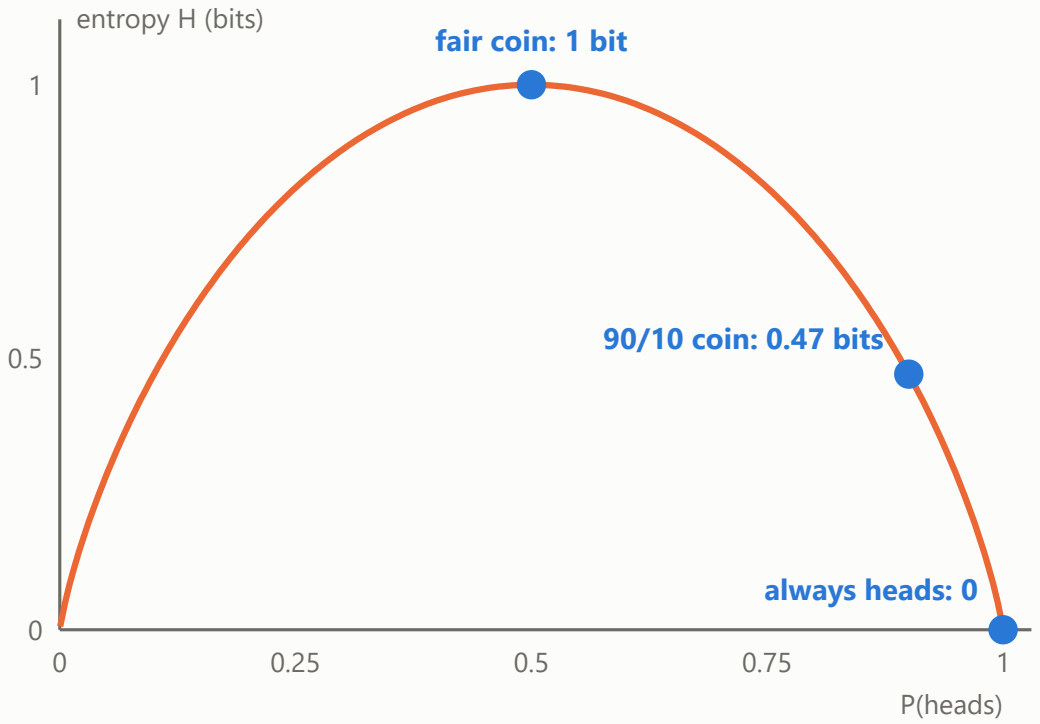


KEY INSIGHT

Halve the probability, add one bit. One fair yes/no answer is exactly 1 bit.

Entropy is the average surprise

- **$H = -\Sigma\ p(x)\ \log_2\ p(x)$**: each outcome's probability × its surprise, added up
- A 90/10 coin: H = 0.9 × 0.152 + 0.1 × 3.32 ≈ **0.47 bits**
- 0 when the outcome is certain; largest when all outcomes are equally likely
- Maximum = $\log_2$(number of outcomes): a fair coin 1 bit, four equal weathers 2 bits



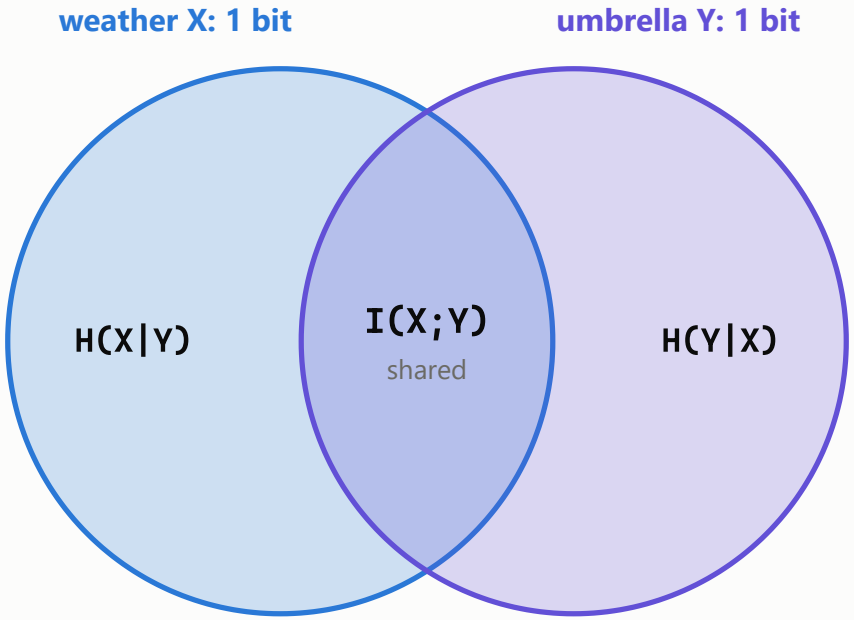
KEY INSIGHT

Entropy is highest when you have no idea what comes next.

Mutual information: what two things share

- **I(X;Y)** = the bits of uncertainty about X that disappear once you know Y
- `I(X;Y) = H(X) + H(Y) − H(X,Y)`

Umbrella matches the weather	I(X;Y)	H(X|Y) left
50% (random)	0	1 bit
80% (usually)	0.28 bits	0.72 bits
100% (always)	1 bit	0



both circles together = H(X,Y)

KEY INSIGHT

0 when the two are unrelated; as large as the entropy itself when one fully decides the other.

PART B

Using information

6 Compression

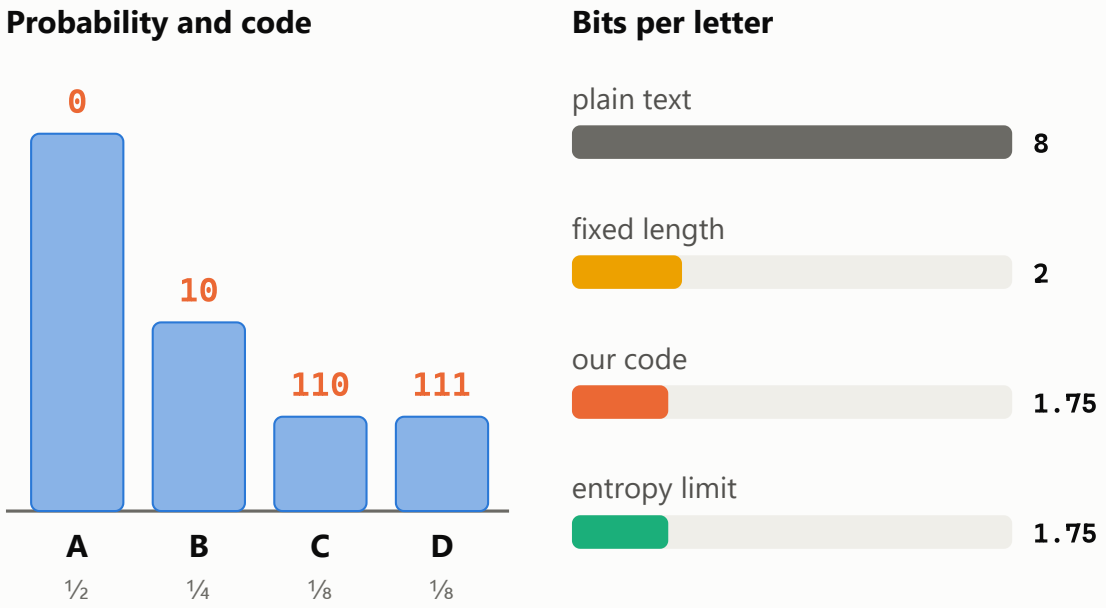
7 Noisy channels

8 Cross-entropy and KL

9 Information theory in AI

Compression gives short codes to common symbols

- Text ABACABAD…: A ½, B ¼, C ⅛, D ⅛ → codes **`A = 0, B = 10, C = 110, D = 111`**
- Average: ½ × 1 + ¼ × 2 + ⅛ × 3 + ⅛ × 3 = **1.75 bits per letter**, instead of 8 (plain text) or 2 (fixed length)
- No code is the start of another, so 010110 can only be A B C



Code length = surprise: 1, 2, 3, 3 bits = $-\log_2(½, ¼, ⅛, ⅛)$

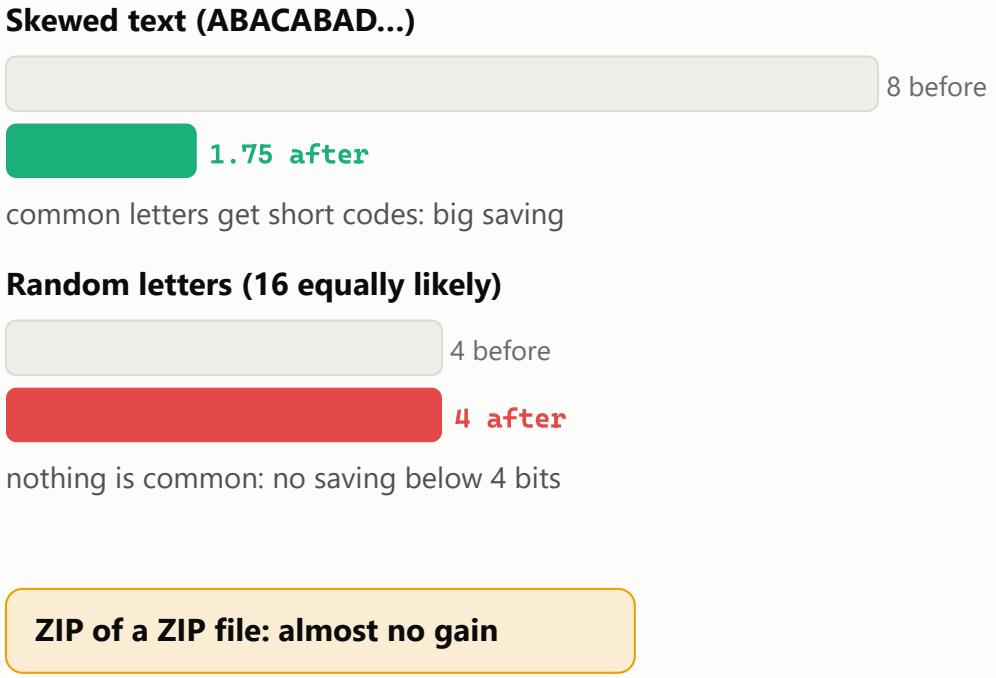
KEY INSIGHT

A good code spends about $-\log_2(p)$ bits on a symbol, so its average length is the entropy: the limit no lossless code can beat.

You can't compress randomness, unless you drop details

Equally likely symbols leave nothing to shorten: 16 equal letters need $\log_2(16) = 4$ bits each; zipping a zip file saves almost nothing.

	Lossless	Lossy
Keeps	every bit, the exact original	only what people notice
Limit	the entropy	far below, by dropping information
Examples	ZIP, PNG, FLAC	JPEG, MP3, streaming video



ZIP of a ZIP file: almost no gain

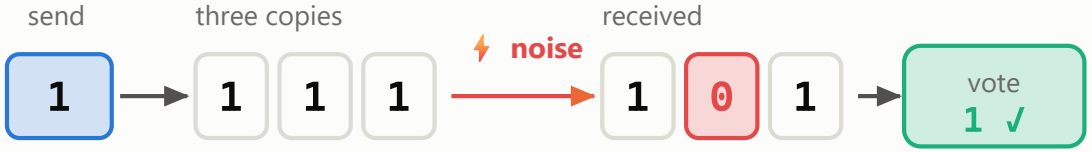
KEY INSIGHT

Lossless compression stops at the entropy; lossy compression goes lower by throwing details away on purpose.

Smart redundancy fights noise

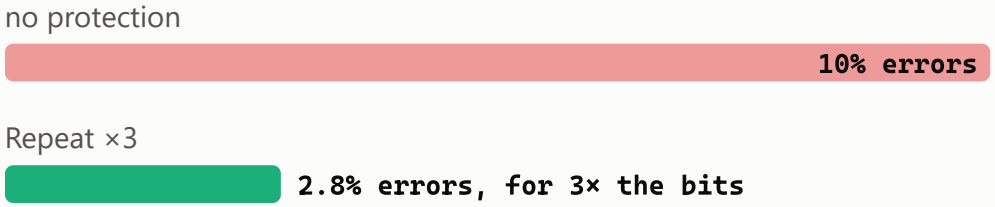
Protection	Extra cost	Detects?	Fixes?
None	0	no	no
Parity bit (1 per 4 bits)	+25%	yes, one flip	no: asks for a resend
Repeat ×3	+200%	yes	yes, by majority vote

With 10% noise, Repeat ×3 is wrong only when 2 or 3 copies flip: **2.8%** instead of 10%.



Wrong only if 2 or 3 copies flip (10% noise):

$$3 \times 0.1 \times 0.1 \times 0.9 + 0.1^3 = 0.028$$

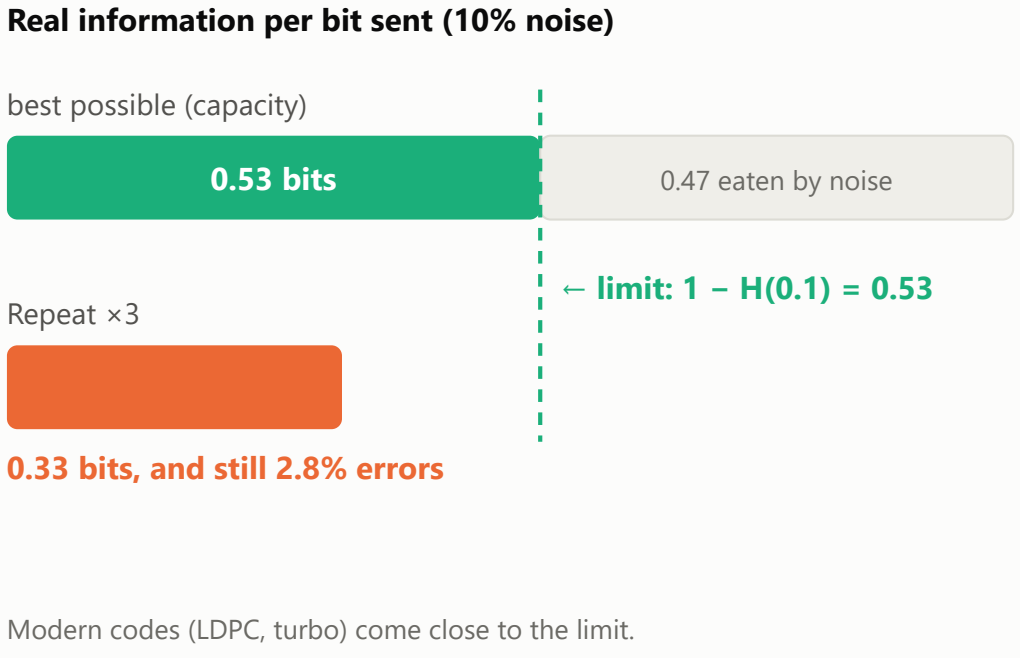


KEY INSIGHT

Useful redundancy (planned check bits) lets the receiver notice, or even fix, flipped bits.

Every channel has a speed limit: its capacity

- **Capacity** = the most information per bit sent that can get through: the largest mutual information between input and output
- Below capacity, smart codes make errors as rare as you like; above it, no code works
- A channel that flips 10% of bits: capacity = 1 − H(0.1) = 1 − 0.47 = **0.53 bits per bit sent**
- Repeat ×3 carries only 0.33 bits per bit sent, and still has 2.8% errors



Modern codes (LDPC, turbo) come close to the limit.

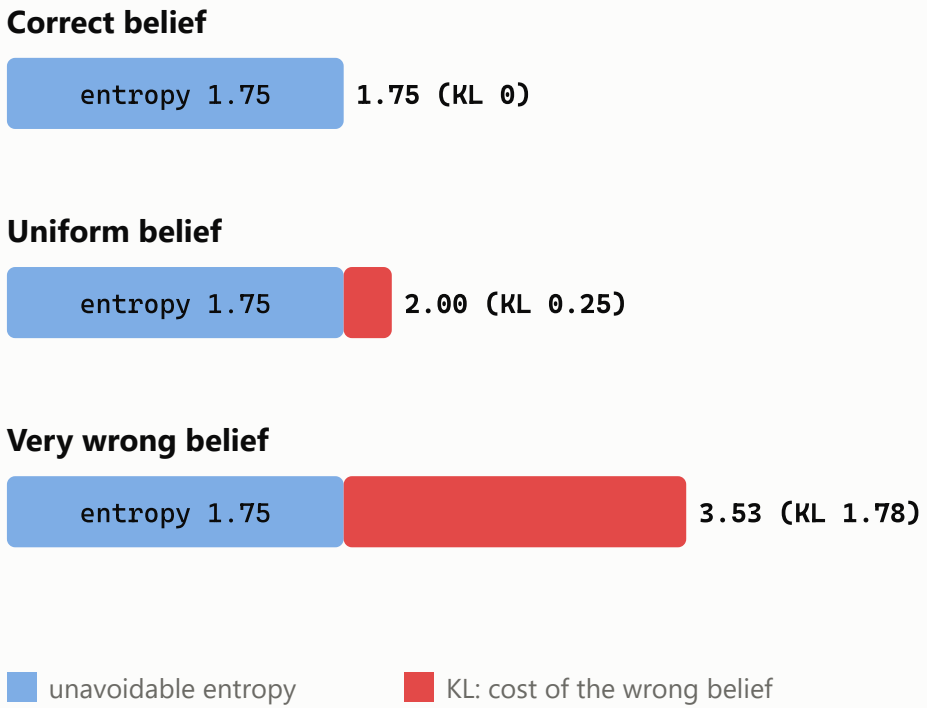
KEY INSIGHT

Shannon proved the limit exists in 1948; it took engineers about 45 years to build codes that come close.

A wrong belief costs extra bits

- **Cross-entropy $H(p, q) = -\Sigma\ p(x)\ \log_2 q(x)$**: your average surprise when the world follows p but you believe q
- **KL(p||q) = H(p, q) − H(p)**: the extra cost, always 0 or more, and 0 only when q = p

Belief about the weather (true H = 1.75)	Cross-entropy	KL
Correct (½, ¼, ⅛, ⅛)	1.75 bits	0
Uniform (¼ each)	2.00 bits	0.25 bits
Very wrong (0.05, 0.05, 0.45, 0.45)	≈ 3.53 bits	≈ 1.78 bits

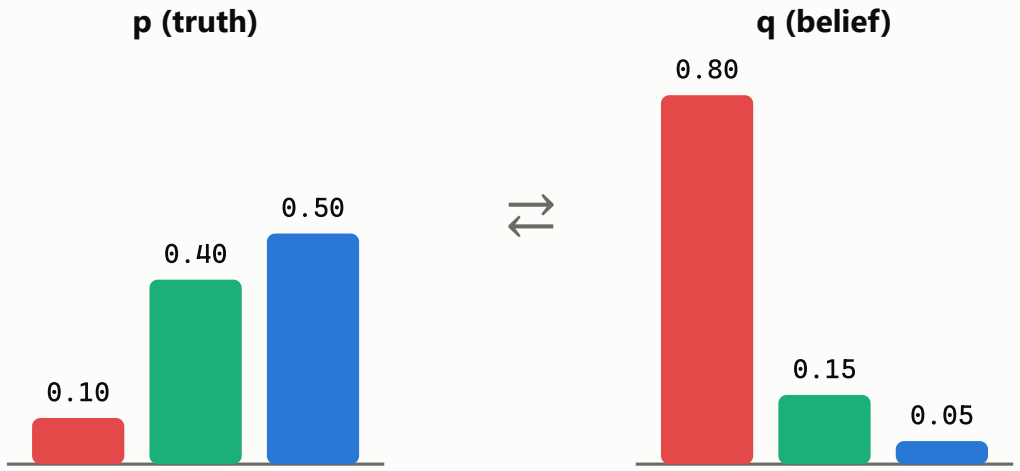


KEY INSIGHT

Cross-entropy = the unavoidable entropy + the KL penalty for believing the wrong thing.

Direction matters: KL is not a distance

- Two distributions over red, green and blue balls: **p = (0.10, 0.40, 0.50)**, **q = (0.80, 0.15, 0.05)**
- H(p, q) = 3.288 bits, but H(q, p) = 2.906 bits
- KL(p||q) ≈ 1.927 bits, but KL(q||p) ≈ 2.022 bits



KL(p||q) ≈ 1.927: blue dominates
q made blue 10× rarer than p

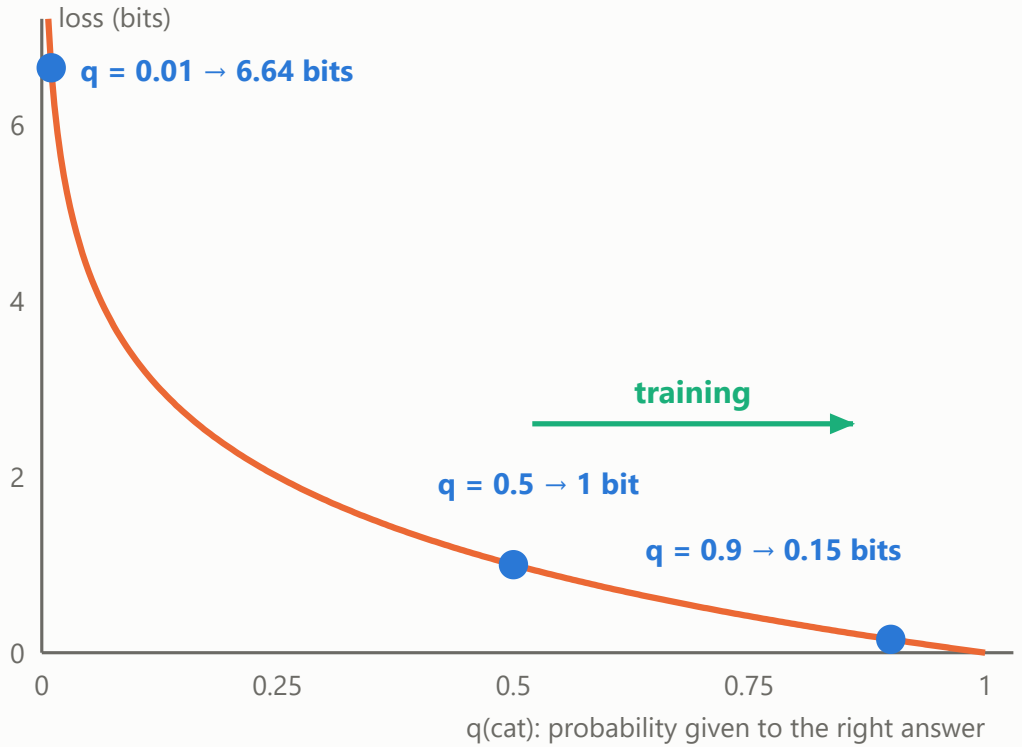
KL(q||p) ≈ 2.022: red dominates
p made red 8× rarer than q

KEY INSIGHT

The weights come from the first distribution, so each direction cares most about the events it thinks are common.

Training a classifier makes the right answer less surprising

- When the truth is "cat", cross-entropy has one term: **`loss = −log₂ q(cat)`**, the surprise of the right answer
- q = 0.9 → 0.15 bits · q = 0.5 → 1 bit · q = 0.01 → 6.64 bits
- ML libraries use the natural log, so the loss is in nats: 1 nat ≈ 1.44 bits

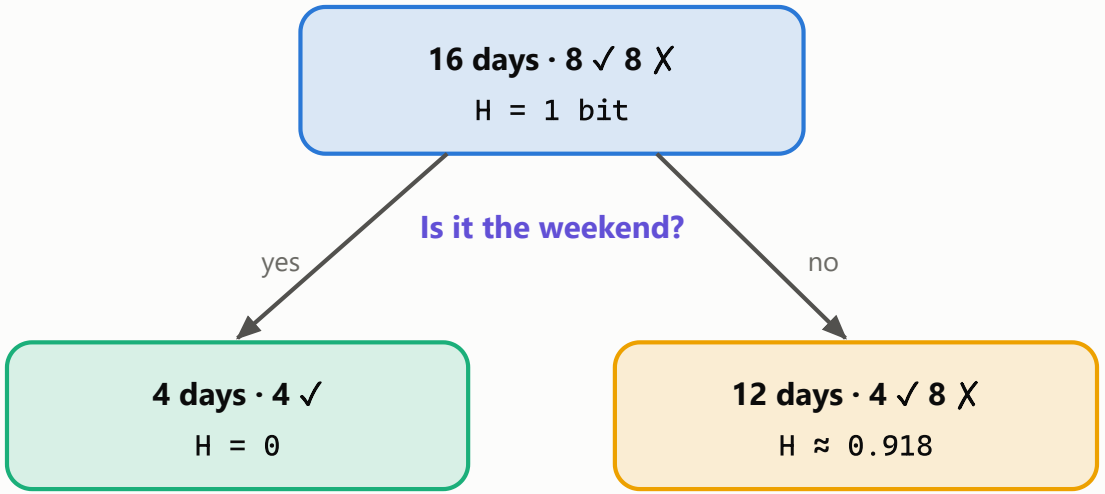


KEY INSIGHT

The loss is the surprise curve from Part A: confident mistakes are punished hardest.

Decision trees ask the most informative question

- **Information gain** = entropy before − average entropy after = the mutual information between the question and the answer
- 16 days, 8 played outside: H = 1 bit
- "Is it the weekend?" → 4 days (all played, H = 0) and 12 days (4 played, H ≈ 0.918) → gain ≈ **0.31 bits**
- "Is it windy?" splits 4/4 and 4/4 → gain 0



`average after = 4/16 × 0 + 12/16 × 0.918 ≈ 0.69`

gain = 1 − 0.69 ≈ 0.31 bits

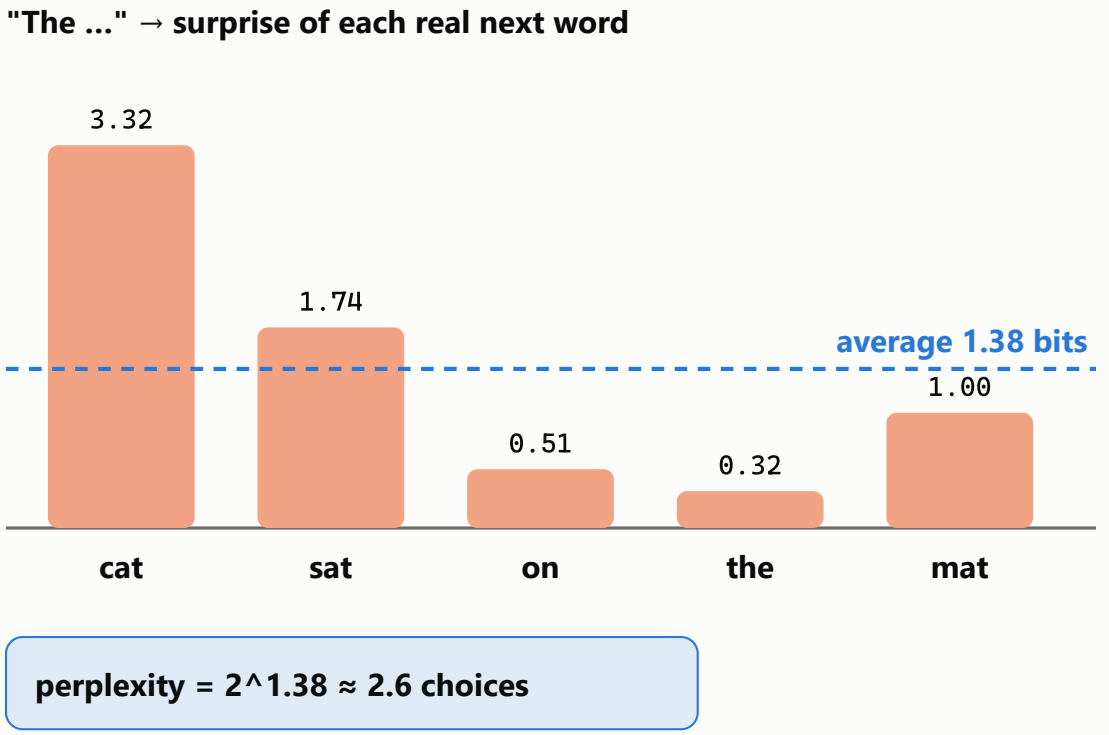
"windy?" gains 0

KEY INSIGHT

The best question is the one that removes the most uncertainty.

Perplexity: how many words a language model is choosing between

- Score a language model on real text: average the surprise of each actual next word
- **`Perplexity = 2^(average surprise)`**
- An easy sentence: average 1.38 bits → perplexity ≈ **2.6**; an odd sentence: 5.81 bits → ≈ **56**

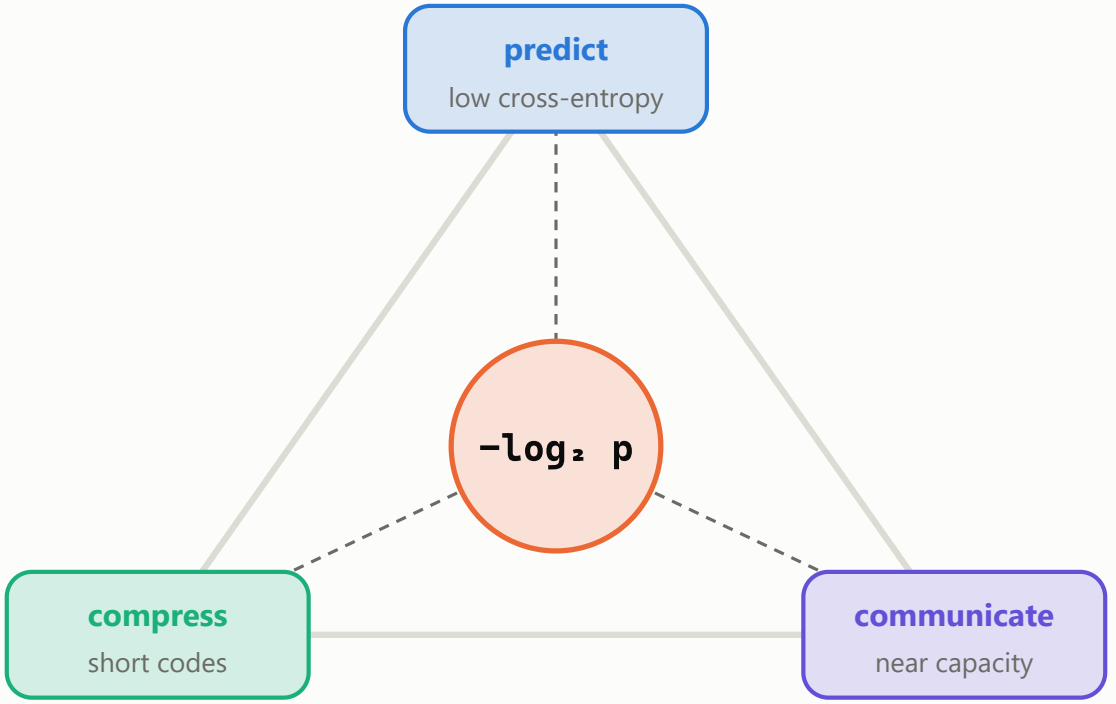


KEY INSIGHT

Lower perplexity = a better model of the language: it is less surprised by real text.

Predicting, compressing and communicating are one problem

Field	Idea	Example
Passwords	surprise	8 random lowercase letters ≈ 37.6 bits; keys use 128 or 256 bits
Biology	entropy	DNA's 4 letters carry at most 2 bits each
Internet	compression + capacity	every file, video call and phone signal



KEY INSIGHT

A model that predicts well is surprised less, compresses well, and needs fewer bits to describe the world.

Key takeaways

1. $\log_2(n)$ counts the yes/no answers needed to pick one of n things.
2. Information is what reduces uncertainty, not what a message means.
3. Surprise = $-\log_2(p)$: rare events carry more bits.
4. Entropy is the average surprise, highest when everything is equally likely.
5. Mutual information is what two things share.
6. Entropy is the limit of lossless compression.
7. Redundancy fights noise; capacity is the speed limit.
8. Cross-entropy = entropy + the KL cost of a wrong belief.
9. AI models are trained and scored by surprise: loss, information gain, perplexity.
10. Predicting, compressing and communicating are one problem.

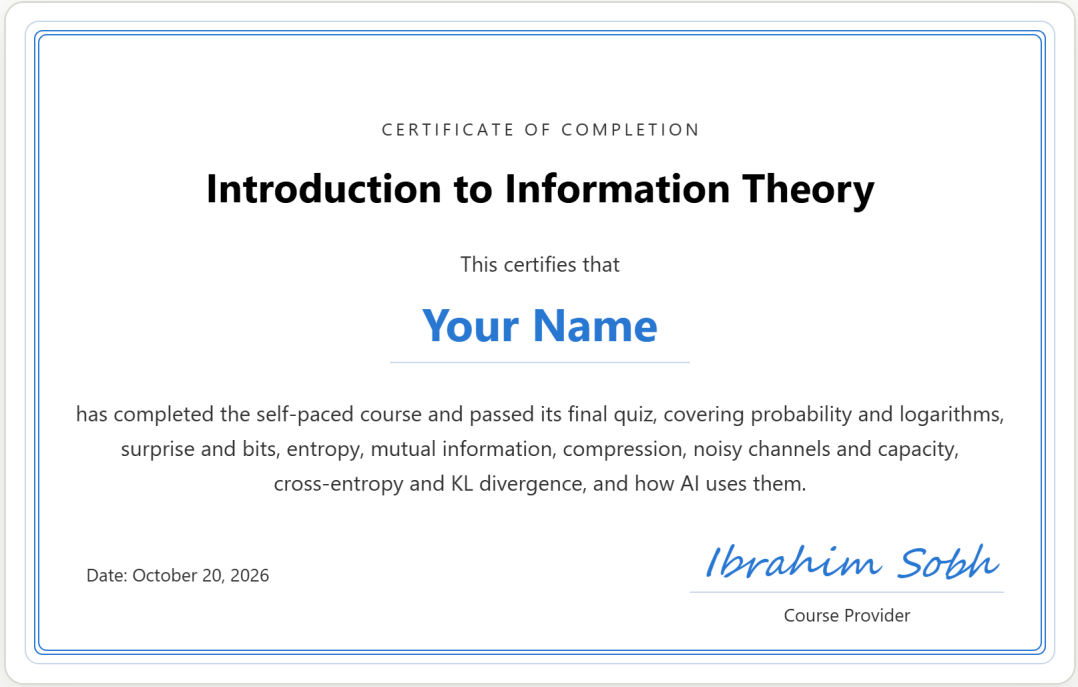
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Introduction to Information Theory: a free, self-paced interactive course

`ibrahimsobh.github.io/introduction-to-information-theory`

- Every idea in this talk has a live demo you can drive: turn events into bits, sweep a coin across the entropy curve, overlap two information circles, compress a text, protect bits from noise, compare a belief with the truth, score a classifier and a language model
- No sign-up, runs in your browser. Pass the final quiz (10 of 12) for a free certificate

Source on GitHub · ⭐ Star the repo if you found it useful.



KEY INSIGHT

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